

December 18, 2025

1. By how much does the emitted current change if the tungsten cathode temperature is increased from $T = 2200$ K to 2350 K? The work function for tungsten is $W_{\text{ki}} = 4.5$ eV. The Richardson coefficient is $6.02 \cdot 10^5$ A/m²K².

Using the Richardson–Dushman equation

$$J = AT^2 \exp\left(-\frac{W}{kT}\right),$$

the ratio of the emitted currents at temperatures $T_1 = 2200$ K and $T_2 = 2350$ K is

$$\frac{J_2}{J_1} = \left(\frac{T_2}{T_1}\right)^2 \exp\left[-W\left(\frac{1}{kT_2} - \frac{1}{kT_1}\right)\right].$$

Substituting $W = 4.5$ eV and $k = 8.617 \times 10^{-5}$ eV/K yields

$$\frac{J_2}{J_1} \approx 5.19.$$

2. What retarding voltage reduces the emission current density to one fifteenth for a cathode at $T = 2200$ K? The electron charge is $q = -1.6 \cdot 10^{-19}$ As.

The retarding voltage reduces the emission current density as

$$J = J_0 \exp\left(-\frac{|q|V}{kT}\right),$$

and we require

$$\frac{J}{J_0} = \frac{1}{15}.$$

Thus

$$V = \frac{kT}{|q|} \ln 15.$$

For $T = 2200$ K,

$$V \approx 0.045 \text{ V}.$$

3. What is the free-electron drift velocity in a copper conductor of radius $R = 1.5$ mm if the current is $I = 12$ A? The electron density in copper is $n_e = 3.3 \cdot 10^{28}$ m⁻³.

The drift velocity in a copper wire is given by

$$v_d = \frac{I}{nqA},$$

where $A = \pi R^2$ with $R = 1.5$ mm and $n = 3.3 \times 10^{28}$ m⁻³. Thus,

$$v_d = \frac{12}{(3.3 \times 10^{28})(1.6 \times 10^{-19})\pi(1.5 \times 10^{-3})^2} \approx 2.04 \times 10^{-4} \text{ m/s}.$$

4. We examine an n-type semiconductor under the conditions shown in the figure: $B_y = 0.4$ T, $U_H = 10$ mV, $I = 30$ mA. The magnetic induction is directed along the y -axis.

(a) What is the electron concentration?

(b) What is the drift velocity of the electrons?

(c) What is the Hall constant?

We examine an n-type semiconductor under the conditions:

$$B_y = 0.4 \text{ T}, \quad U_H = 10 \text{ mV}, \quad I = 30 \text{ mA}.$$

(a) The Hall effect gives

$$U_H = \frac{IB}{nqt},$$

hence the electron concentration

$$n = \frac{IB}{|q|U_H t}.$$

(The thickness t must be known to compute a numerical value.)

(b) The drift velocity is

$$v_d = \frac{I}{nqA},$$

where A is the cross-sectional area of the sample.

(c) The Hall constant is

$$R_H = \frac{1}{nq}.$$